Responses to Reviewers' Comments for Manuscript JSAC-01286-2025

Decentralized Parameter-Free Online Learning

Addressed Comments for Publication to

IEEE Journal on Selected Areas in Communications

by

Tomas Ortega and Hamid Jafarkhani

Dear Dr. Zhiguo Ding,

Please find enclosed the revised version of our previous submission entitled "Decentralized Parameter-Free Online Learning" with manuscript number JSAC-01286-2025. We would like to thank you and the reviewers for the valuable comments which helped improving the quality of our manuscript. In this revision, we have carefully addressed the reviewers' comments. A summary of main modifications and a detailed point-by-point response to the comments from Reviewers 1 to 3 (following the reviewers' order in the decision letter) are given below.

Sincerely,

Tomas Ortega and Hamid Jafarkhani

Note: To enhance the legibility of this response letter, all the editor's and reviewers' comments are typeset in boxes. Rephrased or added sentences are typeset in red. The respective parts in the manuscript are blue to indicate changes.

Contents

Authors' Response to the Editor

> **General Comments.** The manuscript requires a major revision. The main concerns are communication cost in constrained decentralized networks, the gap between the theoretical increasing-gossip schedule and practical constant communication, the role of DECO-ii relative to DECO-i, and the need for stronger empirical comparisons including adaptive baselines and real-data experiments.

Response: We thank the editor and reviewers for the detailed and constructive comments.

We revised the manuscript and repository to address these concerns directly. The main changes are:

1. We made the communication model explicit: agents communicate over a fixed, connected, undirected graph, and each gossip repetition is a local exchange with immediate neighbors. We distinguish DECO-i, which sends (Wealth, G), from DECO-ii, DOGD, and the three adaptive baselines, which send one d-dimensional vector.
2. We state the sufficient schedules precisely as $q(t) = \lceil \kappa t \rceil$, with the potential- and topology-dependent lower bound on κ from each theorem. The practical experiments emphasize the constrained $q(t) = 1$ regime.
3. We reframed DECO-ii as a one-state betting-function formulation rather than an unqualified approximation to DECO-i. The revised manuscript explains that the potential is a certified wealth lower bound and that the variants generally follow different trajectories.
4. We added three representative online decentralized adaptive baselines—D-AdaGrad, D-RMSProp, and D-Adam—with the same gossip interface and communication accounting as DOGD.
5. We evaluated DOGD and all three adaptive baselines over the same 25 learning-rate scales on identical synthetic and LIBSVM online streams. Figs. 3 and 6 disclose

a five-times-loss display mask for extreme values; Figs. 4 and 5 remain focused ablations of connectivity and communication schedule.

6. We revised the experimental discussion to avoid overclaiming empirical dominance: tuned adaptive methods can be competitive or better on some streams, while DECO avoids learning-rate selection and provides explicit communication accounting.
7. We corrected the theory presentation: the reward-regret sign convention and dual-space typing, the radial Hilbert-space lift, the network-regret decomposition, the causal gossip index, the exact exponential-potential normalization, and both schedule constants. Both variants retain the average-local-regret guarantee; the network-regret theorem is stated specifically for DECO-ii.
8. We expanded the related-work discussion to cover recent graph-dependent and communication-efficient decentralized OCO, and sharpened the distinction between those settings and our learning-rate-free, comparator-adaptive guarantees.

For transparency, the relevant added or modified manuscript text supporting every item is reproduced under the corresponding point-by-point response below; no separate manuscript change was made solely for this summary.

Authors' Response to Reviewer 1

General Comments. The paper proposes a family of parameter-free decentralized online learning algorithms based on coin-betting and a proof of the sub linearity of the regret. After reading the paper I have the following comments:

Response: We thank the reviewer for identifying the communication-efficiency issues that needed to be stated more concretely.

Comment 1

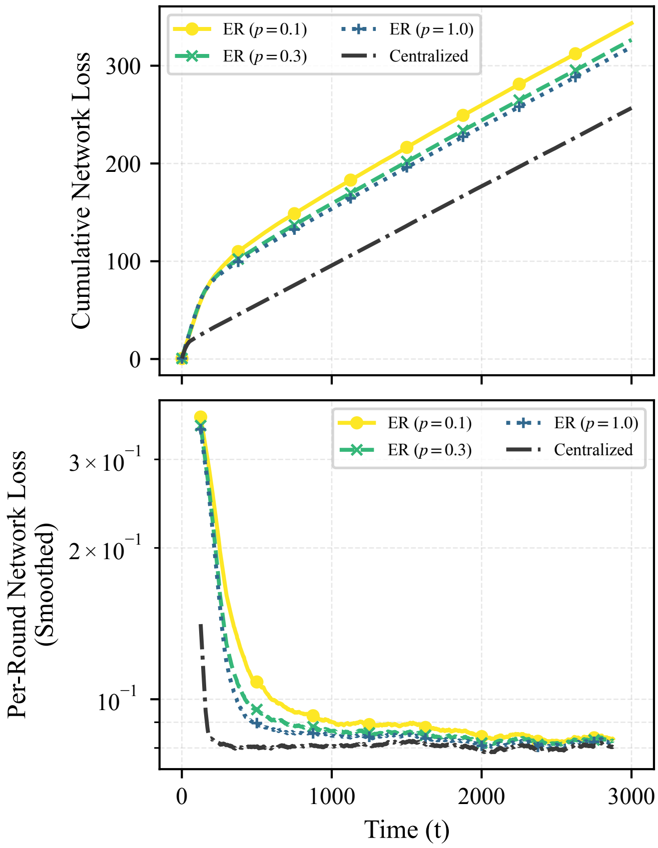
Although the paper proposes an interesting solution backed by regret guarantees, it is not clear for me what is the considered communication setup. In which way does the proposed approach fit the constrained-network scenario of the call.

Response: We appreciate the reviewer's concern, and we revised the manuscript to explicitly describe the communication setup. About the fit with the journal's call for papers, we argue that any "drop-in" online algorithm, which is the case for many IoT applications such as dropping a set of sensors on a field which must operate in a constrained network, requires learning rate tuning, which is particularly challenging given the distributed nature of the problem. The proposed method is parameter-free and does not require learning rate tuning, which is a practical advantage in such scenarios. The manuscript now states:

> We assume synchronous exchanges over a fixed, connected, undirected graph, whose nodes represent the agents.
> At Round t, $q(t)$ repetitions of the fixed gossip matrix are represented by $P_t = W^{q(t)}$. The analysis uses the resulting contraction factor $\rho^{q(t)}$; it does not require treating P_t, whose support may extend beyond immediate neighbors, as a one-round gossip matrix.

DECO-**i** is the direct decentralized analogue of classical wealth-based coin betting: because its decision depends on both $\text{Wealth}_{n,t-1}$ and $G_{n,t-1}$, both states are gossiped. DECO-**ii** is the one-state betting-function variant $x_{n,t} = h_t(G_{n,t-1})$ and gossips only G. Thus, DECO-**i** is a reference construction that preserves the classical wealth recursion, whereas DECO-**ii** reduces each directed-neighbor exchange from $d+1$ to d real scalars, where d is the dimension of the decision space $\mathcal{H}$.

This makes clear that the proposed method is local-neighbor decentralized and identifies the per-message payload of each variant.



Revised manuscript Fig. 4. Synthetic online performance of DECO-**ii** (KT) under varying connectivity, with ER edge probabilities $p \in \{0.1, 0.3, 1.0\}$ and the centralized reference. **Top**: cumulative network loss. **Bottom**: per-round network loss smoothed over 250 rounds.

Comment 2

How is the gossiping happening? What is the communication overhead of the solution? According to the authors $O(t)$ gossiping steps are needed in each round, which means that the gossiping scales with $O(T^2)\ldots$ Going back to point 1, how does this impact its application in constrained networks

Response: We agree that the previous text blurred a sufficient condition for the proof with a practical communication recommendation. We now distinguish the number of gossip repetitions from the payload sent in each neighbor exchange. The sufficient condition is $q(t) = \lceil \kappa t \rceil$, where κ satisfies the potential- and topology-dependent lower bound in the applicable theorem, while the practical communication-constrained regime uses $q(t) = 1$. Thus, the constant gossip uses a linear number of repetitions through horizon T, whereas the sufficient linear schedule uses a quadratic total number. We agree that the linear schedule is impractical in constrained networks, and we do not claim that it is a practical recommendation. Closing the theory/practice gap for $q(t) = \mathcal{O}(1)$ remains an important theoretical question, as we state in the manuscript. The corresponding revised manuscript text is:

The exponential- and KT-potential theorems provide a clear trade-off between communication and performance. They establish that a sufficiently large linear schedule $q(t) = \lceil \kappa t \rceil$, with the potential- and topology-dependent lower bound on κ stated in each theorem, controls the disagreement penalty at order $\sqrt{T}$. The per-round communication is linear and the total communication through horizon T is quadratic.

These coefficient bounds are sufficient under the present disagreement analysis and are not claimed to be necessary. Whether a constant number of gossip rounds, $q(t) = \mathcal{O}(1)$, suffices for sublinear network regret remains open. Accordingly, the constant-communication experiments evaluate the practically constrained regime, while the linear-schedule experiment is a theory-motivated empirical ablation. If

$\rho = 0$, one gossip round gives exact consensus and the logarithmic coefficient bounds are unnecessary.

The scalar count uses the same directed-neighbor accounting for every method: DECO-**i** sends (Wealth, G), while DECO-**ii**, DOGD, and the three adaptive baselines send one d-dimensional vector.

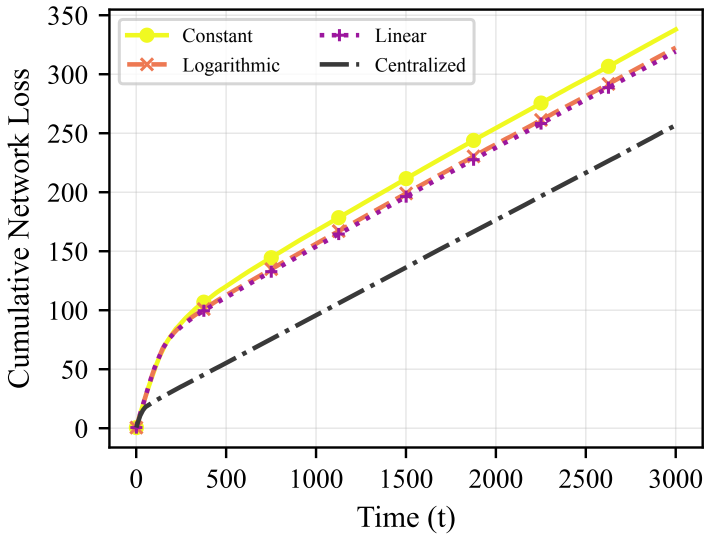
Comment 3

The authors show in Figure 5 the trade-off between communication and performance and claim that "...a linear schedule of the communication allows the decentralized algorithm to closely track the performance of the centralized oracle". However, from the figure one can see that the losses of the linear schedule are approximately 50% higher than the centralized solution. This is not closely tracking, there is a clear gap that the authors should address

Response: By "closely tracking" we meant the slopes are the same. But, we agree that it was misleading. We revised the discussion to treat the centralized gap as a real finite-horizon decentralization cost. The revised discussion is:

For $q(t) = \lceil 0.1t \rceil$, the cumulative number of gossip rounds is quadratic in T, whereas the constant schedule uses only T rounds. The linear schedule reduces loss relative to the constant and logarithmic schedules. While the linear scheme maintains the same slope, a visible finite-horizon gap to the centralized oracle remains. This gap is a real decentralization cost rather than a claim of equality with the oracle. The linear form is theory-motivated, but the empirical coefficient 0.1 is a tractable ablation setting and is not claimed to satisfy the sufficient topology-dependent bound in the KT-potential theorem.

Revised figure caption: Cumulative network loss on the synthetic online stream for DECO-**ii** (KT) with constant, logarithmic, and linear gossip schedules, together with the centralized reference. Increasing the communication budget reduces loss.



Revised manuscript Fig. 5. Cumulative network loss on the synthetic online stream for DECO-**ii** (KT) with constant, logarithmic, and linear gossip schedules, together with the centralized reference. Increasing the communication budget reduces loss.

Comment 4

Intuition into how everything is tighten to a real application is missing

Response: We appreciate that the big idea could use some clarification, and we expanded the operational description in the manuscript to connect the algorithm to distributed sensing and collaborative prediction. The added manuscript text is:

An operational example is collaborative streaming regression over a sensor or device-to-device network. At Round t, each node predicts from its local stream, observes its local feedback, updates its coin-betting state, and gossips the communicated state only with physical or logical neighbors. No central coordinator is required, and the communication budget is determined by these local neighbor exchanges.

Authors' Response to Reviewer 2

General Comments. This paper proposes a decentralized parameter-free online learning framework based on coin-betting and gossip mechanisms, claiming to be the first to achieve network regret guarantees without hyperparameter tuning. While the topic is relevant and potentially impactful, I have several fundamental concerns regarding novelty, technical depth, and empirical validation.

Response: We thank the reviewer for pushing us to sharpen the contribution and empirical claims.

Comment 1

First, the claimed novelty is not sufficiently justified. The proposed method essentially combines existing coin-betting techniques with standard gossip-based consensus schemes. While this integration is non-trivial at an implementation level, the paper does not convincingly argue why this constitutes a conceptual breakthrough rather than a straightforward extension of prior work in decentralized online convex optimization and adaptive methods. The positioning against related literature is incomplete and lacks a clear delineation of what is fundamentally new.

Response: We revised the positioning to be more precise. We acknowledge the recent graph-dependent and communication-efficient D-OCO literature and do not claim novelty for gossip or parameter-free decentralization by themselves. However, to the best of our knowledge, no parameter-free decentralized approach with network-regret guarantees has been proposed for the simultaneous protocol. We identify our contribution as the formulation and analysis that provide both variants of our algorithm average-local-regret guarantees and the second variant a communication-dependent network-regret guarantee. We repeat the relevant manuscript text below:

Decentralized OCO (D-OCO) with every agent active has previously been analyzed using prescribed step-size schedules. Recent global-loss D-OCO analyses make the dependence on topology and communication more explicit. Wan et al. use accelerated gossip to obtain nearly optimal dependence on the horizon, network size, and spectral gap, and their expanded journal work also develops projection-free variants with explicit communication-round guarantees. Lei et al. characterize regret and communication trade-offs over Erdős-Rényi random networks. Communication-efficient D-OCO has also been pursued through block updates with $\mathcal{O}(\sqrt{T})$ communication rounds, difference-compressed messages with explicit transmitted-bit accounting, and multiple primal and gradient consensus iterations for dynamic regret. More recent extensions address unknown feedback delays and compressed communication with improved graph- and compression-dependent regret and corresponding lower bounds. These works establish communication efficiency and graph-dependent regret as central concerns in D-OCO, but address different angles from the learning-rate-free, comparator-adaptive guarantee studied here. Comparator-adaptive distributed learning has also been developed for a protocol in which one agent is activated per round, while stochastic agent availability has been treated separately. These protocols differ from ours, where every agent predicts and incurs a local loss in every round; to the best of our knowledge, no parameter-free decentralized approach with network-regret guarantees has been proposed for this simultaneous protocol.

Parameter-free decentralization itself is not unique to our work. D-NASA targets stationarity for nonconvex stochastic objectives (Li et al.), Kuruzov and coauthors obtain linear convergence for strongly convex smooth objectives, and Chen and coauthors address composite convex objectives. These methods solve fixed-objective optimization problems and measure stationarity or objective convergence, not online optimization. Separately, DADAM is a decentralized adaptive-moment method for online optimization with a dynamic-regret analysis (Nazari et al.), but it retains optimizer hyperparameters. Our contribution instead concerns sequential predict-before-feedback losses: we present two coin-betting algorithm variants, both of which have comparator-adaptive average-local-regret guarantees, and one of which has a communication-dependent network-regret guarantee without a learning-rate scale.

The principal novelty is the decentralized online-learning formulation and analysis that (i) gives both variants average-local-regret guarantees, (ii) upper-bounds the network regret of the second variant by average local coin-betting regret and a communication-controlled disagreement penalty, and (iii) uses its one-state betting-function formulation to obtain that direct network-regret analysis.

Comment 2

Second, the theoretical contribution appears limited in depth. The main regret results largely follow from known reward-regret duality arguments and standard properties of gossip averaging. The decomposition of network regret into local regret plus a disagreement term is expected and has appeared in various forms in prior decentralized optimization literature. The analysis, while technically correct, does not introduce significantly new proof techniques or insights beyond adapting existing frameworks.

Response: We agree that the building blocks are classical, and the revised text is more explicit about this. The contribution is in how these tools are combined for network regret in a decentralized online setting, and in identifying the disagreement that must be controlled when decisions are driven by gossiped cumulative gradients. However, we disagree that the techniques are not significantly new: our betting-function formulation is not present in previous literature, and is the key to the analysis. The betting-function formulation is also a new optimal parameter-free online learning algorithm, even in the single-agent case. We revised the manuscript to clarify the contribution and reproduce the relevant text here:

> The analysis combines two classical ingredients: reward-regret duality for the average local coin-betting term and gossip-mixing bounds for consensus. The decentralized component is their coupling when decisions are generated from gossiped cumulative gradients. In particular, the disagreement term exposes how the betting-function Lipschitz constants, network contraction factor, and gossip schedule jointly affect network regret.
>
> Assume every $l_{n,t}$ is convex and 1-Lipschitz, $\{F_t\}$ is an excellent sequence of coin-betting potentials, and W is a gossip matrix. Let DECO-**ii** use the schedule $q(t)$, define $Q(s,t-1) \coloneqq \sum_{r=s}^{t-1} q(r)$, and suppose h_t is L_{h_t}-Lipschitz on the ball $B_{t-1} \coloneqq \{z \in \mathcal{H} : \|z\| \le t-1\}$. Then DECO-**ii** satisfies, for every $u \in \mathcal{H}$,
>
> $$\begin{aligned} R_T^{\text{net}}(u) &\le R_T^{\text{local}}(u) + \Delta_T \\ &\le F_T^*(\|u\|) + \epsilon + 2\sqrt{N} \sum_{t=1}^{T} L_{h_t} \sum_{s=1}^{t-1} \rho^{Q(s,t-1)}, \end{aligned}$$
>
> where
>
> $$\Delta_T \coloneqq \frac{1}{N^2} \sum_{t=1}^{T} \sum_{n=1}^{N} \sum_{m=1}^{N} \|x_{m,t} - x_{n,t}\| \ge 0.$$
>
> Expanding $l_t = N^{-1} \sum_n l_{n,t}$ gives the exact decomposition
>
> $$\begin{aligned} R_T^{\text{net}}(u) &= R_T^{\text{local}}(u) + D_T, \\ D_T &\coloneqq \frac{1}{N^2} \sum_{t=1}^{T} \sum_{n=1}^{N} \sum_{m=1}^{N} \left(l_{n,t}(x_{m,t}) - l_{n,t}(x_{n,t}) \right). \end{aligned}$$
>
> The residual D_T may have either sign, but 1-Lipschitz continuity gives $D_T \le \Delta_T$. Since $x_{n,t} = h_t(G_{n,t-1})$ and $\|G_{n,t-1}\| \le t-1$, the Lipschitz property, the triangle inequality, and the gossip bound control each pairwise decision difference by
>
> $$\|x_{m,t} - x_{n,t}\| \le 2\sqrt{N} L_{h_t} \sum_{s=1}^{t-1} \rho^{Q(s,t-1)}.$$

Comment 3

Third, there is a notable gap between theory and practice. The theoretical guarantees rely on increasing communication rounds (e.g., $q(t) = O(t)$) to control disagreement, which is impractical in realistic decentralized systems due to communication constraints. However, the experiments are conducted primarily under a constant communication budget, without adequately reconciling this discrepancy. This raises concerns about whether the theoretical results meaningfully explain the empirical performance.

Response: We distinguish the sufficient linear schedule used in the analysis from the constant schedule used in the communication-constrained experiments. The proof uses $q(t) = \lceil \kappa t \rceil$ with a sufficiently large topology-dependent coefficient; this is not presented as a practical recommendation. Closing the theory/practice gap for $q(t) = \mathcal{O}(1)$ remains an important theoretical question. The manuscript now states:

The exponential- and KT-potential theorems provide a clear trade-off between communication and performance. They establish that a sufficiently large linear schedule $q(t) = \lceil \kappa t \rceil$, with the potential- and topology-dependent lower bound on κ stated in each theorem, controls the disagreement penalty at order $\sqrt{T}$. The per-round communication is linear and the total communication through horizon T is quadratic.

These coefficient bounds are sufficient under the present disagreement analysis and are not claimed to be necessary. Whether a constant number of gossip rounds, $q(t) = \mathcal{O}(1)$, suffices for sublinear network regret remains open. Accordingly, the constant-communication experiments evaluate the practically constrained regime, while the linear-schedule experiment is a theory-motivated empirical ablation. If $\rho = 0$, one gossip round gives exact consensus and the logarithmic coefficient bounds are unnecessary.

Comment 4

Finally, the experimental evaluation is not sufficiently convincing. The baselines are limited (primarily DOGD and a centralized oracle), and there is no comparison with more advanced adaptive or parameter-free decentralized methods. Moreover, the experiments focus on relatively standard regression tasks and do not demonstrate clear advantages in more challenging or realistic settings (e.g., heterogeneous data, large-scale networks, or adversarial environments).

Response: We added three representative online decentralized adaptive baselines: D-AdaGrad, D-RMSProp, and D-Adam. They combine the local updates of AdaGrad (Duchi et al.), RMSProp (Tieleman and Hinton), and Adam (Kingma and Ba), respectively, with the same decision-gossip step as the DOGD baseline. Each retains a user-selected base scale, and these are simple decentralized controls rather than implementations of DADAM (Nazari et al.). Every baseline follows the same causal predict-observe-update-gossip protocol and is evaluated on an identical stream. We ran DOGD and all three adaptive methods at the same 25 initial scales from 10^{-3} to 10^{3}. The sensitivity figures use the full evaluated grid and disclose that values above five times the lowest finite loss in a panel are masked for legibility. The connectivity and gossip figures vary only the quantity under study. These additions address the requested breadth through adaptive baselines, heterogeneous synthetic data, and four real online streams; we do not claim that they constitute a separate large-network or adversarial benchmark study. The relevant added or modified manuscript text, including the affected figure captions, is reproduced below in manuscript order:

> We compare them against the following baselines:
>
> **Online decentralized adaptive baselines**: D-AdaGrad, D-RMSProp, and D-Adam combine the local diagonal AdaGrad (Duchi et al.), RMSProp (Tieleman and Hinton), and Adam (Kingma and Ba) updates, respectively, with the same

decision-gossip step as DOGD. They retain a user-selected base scale η_0 and are simple experimental controls, not DADAM (Nazari et al.).

All baselines use the same causal online protocol: predict, observe one local gradient, update the local optimizer state, and then gossip the decision vector once unless another schedule is stated. We sweep the same 25 values $\eta_0 \in [10^{-3}, 10^3]$ for DOGD and the three adaptive methods on identical data streams. DOGD uses $\eta_t = \eta_0/\sqrt{t}$, whereas η_0 is the base scale multiplying the online moment-normalized update for D-AdaGrad, D-RMSProp, and D-Adam. All 25 settings are evaluated. For legibility on a common logarithmic scale, a plotted value above five times the lowest finite cumulative loss in its panel is masked; the underlying sweep result is retained. The sensitivity figures therefore do not portray the adaptive methods as parameter-free. The scalar count uses the same directed-neighbor accounting for every method: DECO-**i** sends (Wealth, G), while DECO-**ii**, DOGD, and the three adaptive baselines send one d-dimensional vector.

Final cumulative network loss on the synthetic online stream across the 25-value base-rate grid $\eta_0 \in [10^{-3}, 10^3]$. Values above five times the lowest finite loss are omitted from the plot to keep the useful range legible. Horizontal lines are the untuned DECO-**i**/**ii** (KT) and centralized references.

A primary motivation for our work is the brittleness of online methods that require a user-chosen learning-rate scale. We therefore sweep DOGD, D-AdaGrad, D-RMSProp, and D-Adam over the same 25 values from 10^{-3} to 10^3 on the same synthetic stream for $T = 3000$ rounds. All four baselines operate sequentially without replaying future losses.

As shown in Fig. 3, every baseline has a restricted useful range of η_0, and their losses deteriorate substantially outside it. The optimal scale also differs across DOGD, D-AdaGrad, D-RMSProp, and D-Adam. The horizontal DECO curves require no learning-rate sweep; this is the specific parameter-free advantage supported by the experiment.

Since the KT and exponential potentials provide similar performance, the two sensitivity figures show the KT potential for DECO and the centralized reference to keep their legends readable.

Synthetic online performance of DECO-**ii** (KT) under varying connectivity, with ER edge probabilities $p \in \{0.1, 0.3, 1.0\}$ and the centralized reference. **Top**: cumulative network loss. **Bottom**: per-round network loss smoothed over 250 rounds.

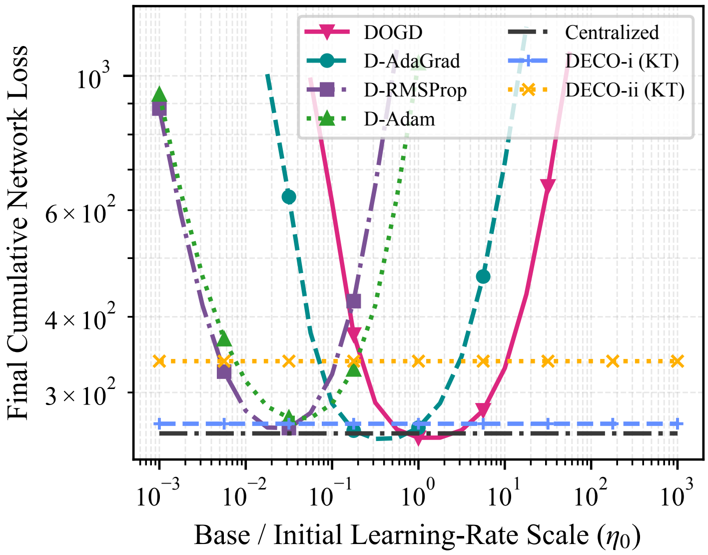
Cumulative network loss on the synthetic online stream for DECO-**ii** (KT) with constant, logarithmic, and linear gossip schedules, together with the centralized reference. Increasing the communication budget reduces loss.

Final cumulative network loss on four online LIBSVM streams across the 25-value η_0 grid. In each panel, values above five times its lowest finite loss are omitted to keep the useful range legible. Horizontal lines show DECO-**i**/**ii** (KT) and the centralized reference without learning-rate tuning.

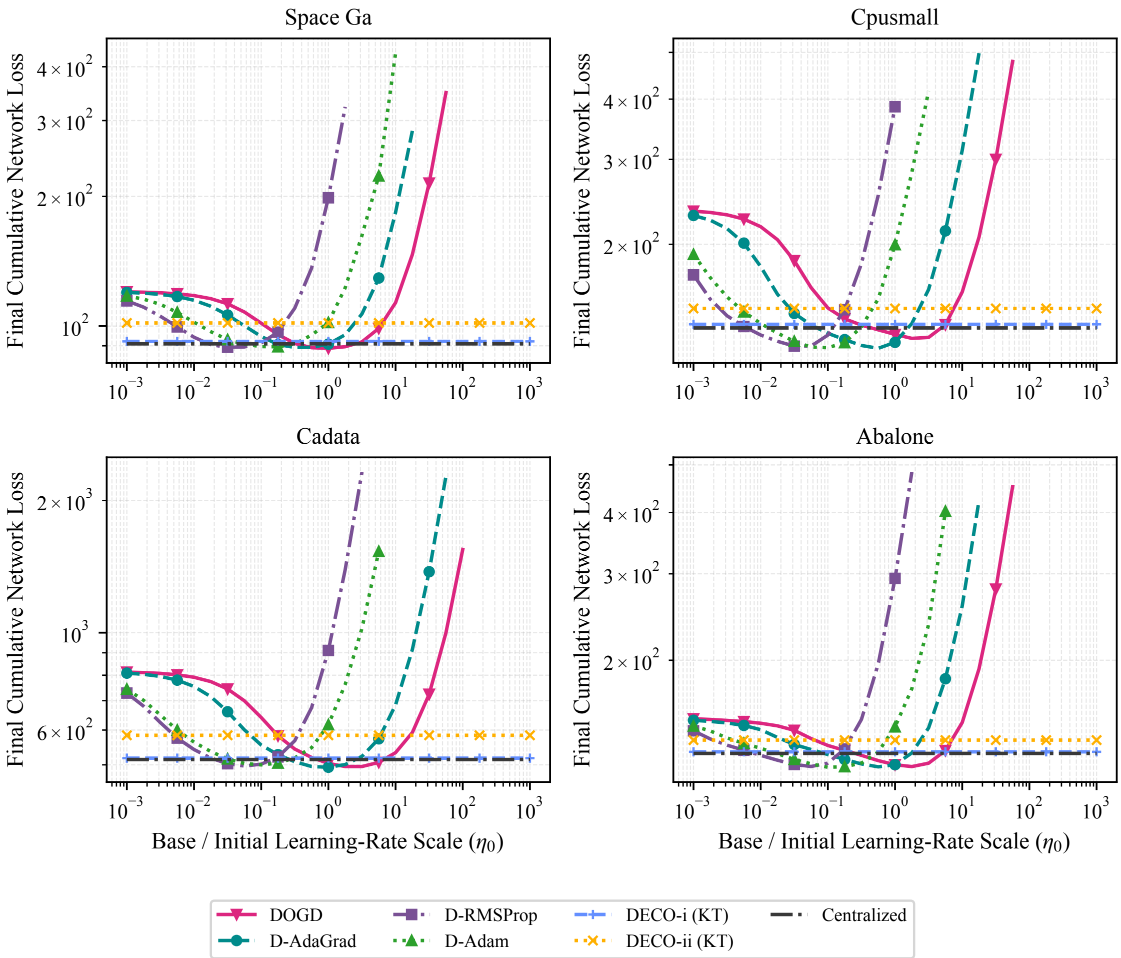
We evaluate DOGD, D-AdaGrad, D-RMSProp, and D-Adam on the same four LIBSVM online streams as DECO. Every run uses a cycle graph with $N = 20$ and $q(t) = 1$, and every baseline is swept over the same 25-value grid $\eta_0 \in [10^{-3}, 10^{3}]$. Fig. 6 uses the entire evaluated grid rather than selecting a best value after the fact; the disclosed five-times-loss mask only affects display.

Fig. 6 shows that learning-rate sensitivity is not specific to DOGD. D-AdaGrad, D-RMSProp, and D-Adam also deteriorate outside method- and dataset-dependent ranges, even though their moment accumulators adapt the coordinate-wise update magnitudes online. Within their favorable ranges they can match or outperform DECO, but the favorable scale varies by method and dataset. Thus, we emphasize that the practical advantage of DECO is its parameter-free nature, which removes the need for learning-rate selection, and maintains competitive performance across a variety of online streams.

The results compare DECO with tuned DOGD, D-AdaGrad, D-RMSProp, and D-Adam across synthetic and real online streams. The tuned baselines can be better on some streams, while DECO remains competitive without selecting a learning-rate scale.



Revised manuscript Fig. 3. Final cumulative network loss on the synthetic online stream across the 25-value base-rate grid $\eta_0 \in [10^{-3}, 10^3]$. Values above five times the lowest finite loss are omitted from the plot to keep the useful range legible. Horizontal lines are the untuned DECO-**i**/**ii** (KT) and centralized references.



Revised manuscript Fig. 6. Final cumulative network loss on four online LIBSVM streams across the 25-value η_0 grid. In each panel, values above five times its lowest finite loss are omitted to keep the useful range legible. Horizontal lines show DECO-**i**/**ii** (KT) and the centralized reference without learning-rate tuning.

Authors' Response to Reviewer 3

General Comments. This paper proposes a coin-betting-based decentralized online learning, DECO, for parameter-free updates with regret guarantees. In particular, the paper presents DECO-i and its simplified variant DECO-ii as the main algorithms.

However, the central claims do not yet appear to be fully supported. Although DECO-ii is presented as the main proposed method, the paper lacks sufficient quantitative analysis of the approximation that it introduces relative to DECO-i, as well as its impact on regret over T iterations. In addition, while the paper emphasizes that the method is learning-rate-free, it still depends on the choice of a potential function, which is itself a consequential design decision. As such, the approach does not fully eliminate the need for tuning, but rather shifts it to another component. Moreover, although DECO-ii has a practical advantage over DECO-i by avoidng communication of the wealth variable and only gossiping the accumulated gradient, the paper does not sufficiently justify why the two-variable communication structure in DECO-i should be considered as an appropriate design in the first place. In many decentralized learning frameworks, communication typically involves a single optimization-related state, such as the model parameters or gradient information. In contrast, DECO-i requires transmitting both the accumulated gradient and the wealth variable at each round, which may introduce unnecessary overhead without sufficient justification. More detailed concerns are as follows:

Response: We thank the reviewer for the detailed technical comments. The reviewer has a list of detailed comments that represent the above concerns. In what follows, we address all reviewer's comments in detail.

Comment 1

The paper suffers from inconsistent notation, which significantly hinders readability and makes the technical development difficult to follow. For example, the subgradient is denoted by g, while $c = -g$ is introduced subsequently. In the regret analysis, both c and g appear interchangeably, which leads to unnecessary confusion. Moreover, the symbol c appearing in Eqs. (57) and (66) no longer represents the coin-flipping outcome introduced earlier, but is instead used with a different meaning. Such overloading of notation substantially reduces the clarity of the paper

Response: We revised the notation discussion to reduce overloading. The manuscript now emphasizes $g_{n,t}$ as the subgradient feedback and uses the identification $c_t = -g_t$ locally when recalling the classical single-agent coin-betting view. We also avoid reusing c for unrelated constants in the disagreement analysis and corrected the operational update to match the algorithm's $G - g$ convention. This makes the OCO and coin-betting interpretations easier to follow. The relevant added notation text and corrected operational equation are:

To keep the two viewpoints distinct, $g_{n,t}$ denotes subgradient feedback throughout the OCO and network-regret development, whereas $c_t = -g_t$ is used only in the local recap of the classical coin-betting game. The coefficient of a linear gossip schedule is denoted by κ, so $q(t) = \lceil \kappa t \rceil$; it is not a coin outcome.

$$\hat{G}_{n,t} = G_{n,t-1} - g_{n,t}.$$

This notation cleanup also prompted us to recheck the causal timing and constants in the schedule proofs. Since $x_{n,t}$ is chosen from $G_{n,t-1}$, round-t disagreement depends on gossip only through round $t-1$. We corrected the index and the resulting KT coefficient, rather than treating this as only a symbol rename. The revised core calculation is:

Define $S_1 = 0$ and

$$S_t = \sum_{s=1}^{t-1} \rho^{Q(s,t-1)}, \qquad Q(s,t-1) = \sum_{r=s}^{t-1} q(r).$$

For $q(r) = \lceil \kappa r \rceil$ and $t \geq 2$,

$$Q(s,t-1) \geq \kappa \sum_{r=s}^{t-1} r = \kappa \frac{(t-s)(s+t-1)}{2} \geq \kappa(t-1),$$
$$S_t \leq (t-1)\rho^{\kappa(t-1)}.$$

The exponential-potential theorem uses

$$\kappa \geq -\frac{3}{2\ln\rho},$$

while the KT theorem uses the corrected sufficient condition

$$\kappa \geq -\frac{4\ln 2}{\ln\rho}.$$

For the exponential potential, $H_t = \sum_{i=1}^{t} i^{-1}$, with $H_0 = 0$ and $F_0(0) = \epsilon$; this is the exact normal-potential normalization from Orabona and Pál.

$$F_0(0) = \epsilon, \qquad F_t(x) = \epsilon \exp\left(\frac{x^2}{2t} - \frac{H_t}{2}\right), \quad t \geq 1.$$

Comment 2

The main proposed method is DECO-ii, whereas DECO-i is essentially the standard coin-betting approach augmented with a gossip step, as also acknowledged by the authors. DECO-ii appears to be obtained by replacing Wealth_{t-1} in DECO-i with the approximation $\mathrm{Wealth}_{t-1} \approx F_{t-1}(G_{t-1})$. The authors should provide a quantitative analysis of the approximation error introduced by this simplification, as well as its impact on regret after T iterations. However, the paper does not provide such an explanation or theoretical characterization. Instead, it only shows experimentally that DECO-ii performs worse than DECO-i, which is insufficient. A rigorous quantitative analysis of the approximation error in DECO-ii and its effect on the regret bound is necessary to substantiate the validity of the method.

Response: We clarify that DECO-ii is not obtained by assuming that the potential approximates wealth. The wealth lemma certifies the potential as a lower bound for DECO-i, while the different local and gossip updates generally lead the two variants along different trajectories. We therefore analyze DECO-ii directly rather than introducing a cumulative approximation-error term. The manuscript now states:

> DECO-**i** is the direct decentralized analogue of classical wealth-based coin betting: because its decision depends on both $\mathrm{Wealth}_{n,t-1}$ and $G_{n,t-1}$, both states are gossiped. DECO-**ii** is the one-state betting-function variant $x_{n,t} = h_t(G_{n,t-1})$ and gossips only G. Thus, DECO-**i** is a reference construction that preserves the classical wealth recursion, whereas DECO-**ii** reduces each directed-neighbor exchange from $d+1$ to d real scalars, where d is the dimension of the decision space $\mathcal{H}$.
>
> The variants are algebraically related: the wealth lemma shows that the potential is a certified wealth lower bound, not an estimator with an assumed approximation accuracy. After the different local and gossip updates, the two variants generally

follow different state trajectories, so DECO-**ii** is analyzed directly rather than through a cumulative approximation-error claim.

Comment 3

Lemma 3 only establishes whether Eq. (4) is satisfied when learning is based on DECO-i. Since DECO-i is essentially corresponds to the conventional coin-betting framework combined with gossip, it is not surprising that the condition holds. However, the paper does not provide a proper analysis to verify whether the same condition is satisfied for DECO-ii, which is the simplified version of DECO-i. Without such an analysis, it remains unclear whether DECO-ii can truly enjoy the same regret guarantees as DECO-i.

Response: We added a direct verification for the betting-function update and corrected the Hilbert-space step. The scalar potential condition cannot be applied componentwise to a vector state; the revised proof uses the standard radial lift of an excellent potential, including condition (d). The radial one-step inequality is then applied before gossip, followed by convexity, nonnegative doubly stochastic mixing, and telescoping. We also made the scalar conjugate type-correct by evaluating it at $\|u\|$. The corresponding revised argument is:

Let V and V^* be a pair of dual vector spaces. Let $F : V^* \to \mathbb{R} \cup \{+\infty\}$ be a proper, convex, lower semi-continuous function and let $F^* : V \to \mathbb{R} \cup \{+\infty\}$ be its Fenchel conjugate. Let $x_1, \ldots, x_T \in V$ and $g_1, \ldots, g_T \in V^*$. For any $\epsilon > 0$, the reward-regret relationship is

$$-\sum_{t=1}^{T}\langle g_t, x_t\rangle \geq F\left(-\sum_{t=1}^{T} g_t\right) - \epsilon$$

if and only if, for every $u \in V$,

$$\sum_{t=1}^{T}\langle g_t, x_t - u\rangle \le F^*(u) + \epsilon.$$

Define the radial potential and betting fraction by

$$\begin{aligned}
\Phi_t(z) &= F_t(\|z\|),\\
\boldsymbol{\beta}_t(z) &= \begin{cases} \beta_t(\|z\|)\dfrac{z}{\|z\|}, & z \neq 0,\\[2ex] 0, & z = 0,\end{cases}\\
h_t(z) &= F_{t-1}(\|z\|)\boldsymbol{\beta}_t(z).
\end{aligned}$$

For an excellent potential and $\|z\| \le t-1$, $\|c\| \le 1$, the radial one-step inequality is

$$\Phi_{t-1}(z) + \langle c, h_t(z)\rangle \ge \Phi_t(z+c).$$

Moreover, Φ_t is convex. Applying it to DECO-**ii** with $c = -g_{n,t}$ gives

$$\Phi_{t-1}(G_{n,t-1}) - \langle g_{n,t}, x_{n,t}\rangle \ge \Phi_t(\widehat{G}_{n,t}).$$

Convexity of Φ_t and the gossip update give

$$\begin{aligned}
\Phi_t(G_{n,t}) &\le \sum_m (P_t)_{nm}\Phi_t(\widehat{G}_{m,t})\\
&\le \sum_m (P_t)_{nm}\left(\Phi_{t-1}(G_{m,t-1}) - \langle g_{m,t}, x_{m,t}\rangle\right).
\end{aligned}$$

Summing over agents, using double stochasticity, and telescoping yields

$$\begin{aligned}
\frac{1}{N}\sum_{t=1}^{T}\sum_{n=1}^{N}\langle g_{n,t}, x_{n,t}\rangle &\le \epsilon - \frac{1}{N}\sum_{n=1}^{N}\Phi_T(G_{n,T})\\
&\le \epsilon - \Phi_T\left(\frac{1}{N}\sum_{n=1}^{N}G_{n,T}\right)\\
&= \epsilon - F_T\left(\left\|\sum_{t=1}^{T}\bar{g}_t\right\|\right).
\end{aligned}$$

Thus, DECO-**ii** satisfies the potential inequality needed by reward-regret duality without storing or gossiping a wealth variable. Together with the wealth lower bound for DECO-**i**, this shows that both variants satisfy

$$R_T^{\mathrm{local}}(u) \le \Phi_T^*(u) + \epsilon = F_T^*(\|u\|) + \epsilon.$$

Comment 4

The authors claim that DECO has the advantage of not requiring learning rate tuning, unlike conventional online learning methods. However, while DECO removes the need to tune a learning rate, it instead requires storing the cumulative sum of past gradients and choosing a potential function, both of which can significantly affect performance. In other words, although one hyperparameter is removed, another design choice, the potential function, is introduced. As a result, the practical advantage of using this algorithm over existing methods is not sufficiently justified.

Response: We agree and revised the language to avoid implying that all design choices disappear. The method is learning-rate-free and comparator- and horizon-adaptive, but the potential function is still a modeling choice. We now state this explicitly and report both KT and exponential potentials. We also clarify that cumulative-gradient storage is one d-dimensional vector per agent, comparable to storing optimizer state in adaptive-gradient methods. The added manuscript text is:

> Here parameter-free means learning-rate-free and comparator- and horizon-adaptive; it does not mean free of every design choice. The potential family remains a modeling choice, and we evaluate both KT and exponential potentials. For the exponential potential, $H_t = \sum_{i=1}^{t} i^{-1}$, with $H_0 = 0$ and $F_0(0) = \epsilon$; this is the exact normal-potential normalization from Orabona and Pál. Each DECO-**ii** agent stores one d-dimensional accumulated-gradient vector; DECO-**i** additionally stores one scalar wealth. This memory is comparable to the vector state used by adaptive-gradient optimizers.

Comment 5

The update magnitude in DECO is ultimately determined by β_t, which itself depends on the cumulative sum of past gradients. Therefore, it is not sufficient to compare DECO only against DOGD, whose learning rate decreases deterministically with t. A more appropriate evaluation would include comparisons with methods that also adapt their effective update size based on past gradients, such as AdaGrad, RMSProp, Adam, AdamW, momentum, and Nesterov-type methods applied in the decentralized online learning setting. Without such baselines, the empirical evaluation does not convincingly demonstrate the claimed advantage of DECO.

Response: Per reviewer's suggestion, we have selected D-AdaGrad, D-RMSProp, and D-Adam as three representative adaptive-gradient baselines. They derive their local updates from diagonal AdaGrad (Duchi et al.), RMSProp (Tieleman and Hinton), and Adam (Kingma and Ba), respectively. AdamW mainly adds weight decay to Adam, while momentum and Nesterov are not adaptive-gradient methods; these three representatives directly test the reviewer's adaptive-method concern without overloading the figures. We implemented each method as a genuinely online decentralized algorithm: at every round it predicts, observes only the current local gradient, updates its local optimizer state, and gossips one decision vector. DOGD and all three adaptive methods use the same 25-value initial-rate grid and identical streams. Figs. 3 and 6 use the full evaluated grid with the disclosed five-times-loss display mask. Figures 4 and 5 are focused controlled ablations of connectivity and gossip schedule, respectively. The revised sensitivity figures are reproduced above with our response to the experimental-evaluation comment. The relevant added or modified manuscript text, including the affected figure captions, is:

We compare them against the following baselines:

Online decentralized adaptive baselines: D-AdaGrad, D-RMSProp, and D-Adam combine the local diagonal AdaGrad (Duchi et al.), RMSProp (Tieleman and Hinton), and Adam (Kingma and Ba) updates, respectively, with the same decision-gossip step as DOGD. They retain a user-selected base scale η_0 and are simple experimental controls, not DADAM (Nazari et al.).

All baselines use the same causal online protocol: predict, observe one local gradient, update the local optimizer state, and then gossip the decision vector once unless another schedule is stated. We sweep the same 25 values $\eta_0 \in [10^{-3}, 10^3]$ for DOGD and the three adaptive methods on identical data streams. DOGD uses $\eta_t = \eta_0/\sqrt{t}$, whereas η_0 is the base scale multiplying the online moment-normalized update for D-AdaGrad, D-RMSProp, and D-Adam. All 25 settings are evaluated. For legibility on a common logarithmic scale, a plotted value above five times the lowest finite cumulative loss in its panel is masked; the underlying sweep result is retained. The sensitivity figures therefore do not portray the adaptive methods as parameter-free. The scalar count uses the same directed-neighbor accounting for every method: DECO-**i** sends (Wealth, G), while DECO-**ii**, DOGD, and the three adaptive baselines send one d-dimensional vector.

Final cumulative network loss on the synthetic online stream across the 25-value base-rate grid $\eta_0 \in [10^{-3}, 10^3]$. Values above five times the lowest finite loss are omitted from the plot to keep the useful range legible. Horizontal lines are the untuned DECO-**i**/**ii** (KT) and centralized references.

A primary motivation for our work is the brittleness of online methods that require a user-chosen learning-rate scale. We therefore sweep DOGD, D-AdaGrad, D-RMSProp, and D-Adam over the same 25 values from 10^{-3} to 10^3 on the same synthetic stream for $T = 3000$ rounds. All four baselines operate sequentially without replaying future losses.

As shown in Fig. 3, every baseline has a restricted useful range of η_0, and their losses deteriorate substantially outside it. The optimal scale also differs across DOGD, D-AdaGrad, D-RMSProp, and D-Adam. The horizontal DECO curves require no learning-rate sweep; this is the specific parameter-free advantage supported by the experiment.

Since the KT and exponential potentials provide similar performance, the two sensitivity figures show the KT potential for DECO and the centralized reference to keep their legends readable.

Synthetic online performance of DECO-**ii** (KT) under varying connectivity, with ER edge probabilities $p \in \{0.1, 0.3, 1.0\}$ and the centralized reference. **Top**: cumulative network loss. **Bottom**: per-round network loss smoothed over 250 rounds.

Cumulative network loss on the synthetic online stream for DECO-**ii** (KT) with constant, logarithmic, and linear gossip schedules, together with the centralized reference. Increasing the communication budget reduces loss.

Final cumulative network loss on four online LIBSVM streams across the 25-value η_0 grid. In each panel, values above five times its lowest finite loss are omitted to keep the useful range legible. Horizontal lines show DECO-**i**/**ii** (KT) and the centralized reference without learning-rate tuning.

We evaluate DOGD, D-AdaGrad, D-RMSProp, and D-Adam on the same four LIBSVM online streams as DECO. Every run uses a cycle graph with $N = 20$ and $q(t) = 1$, and every baseline is swept over the same 25-value grid $\eta_0 \in [10^{-3}, 10^{3}]$. Fig. 6 uses the entire evaluated grid rather than selecting a best value after the fact; the disclosed five-times-loss mask only affects display.

Fig. 6 shows that learning-rate sensitivity is not specific to DOGD. D-AdaGrad, D-RMSProp, and D-Adam also deteriorate outside method- and dataset-dependent ranges, even though their moment accumulators adapt the coordinate-wise update magnitudes online. Within their favorable ranges they can match or outperform DECO, but the favorable scale varies by method and dataset. Thus, we emphasize that the practical advantage of DECO is its parameter-free nature, which removes the need for learning-rate selection, and maintains competitive performance across a variety of online streams.

The results compare DECO with tuned DOGD, D-AdaGrad, D-RMSProp, and D-Adam across synthetic and real online streams. The tuned baselines can be better on some streams, while DECO remains competitive without selecting a learning-rate scale.

Comment 6

The discussion of prior works on parameter-free decentralized learning is insufficient, and the experimental baselines are incomplete in this regard. Several studiess on parameter-free decentralized learning already exit (e.g., [1]–[3]). However, the paper neither provide adequate discussion about these works adequately nor includes relevant parameter-free decentralized methods in the baseline comparisons.

[1] Li, Jiaxiang, et al. "Problem-parameter-free decentralized nonconvex stochastic optimization." arXiv preprint arXiv:2402.08821 (2024).

[2] Kuruzov, Ilya, Gesualdo Scutari, and Alexander Gasnikov. "Achieving linear convergence with parameter-free algorithms in decentralized optimization." Advances in Neural Information Processing Systems 37 (2024): 96011–96044.

[3] Chen, Xiaokai, et al. "A parameter-free decentralized algorithm for composite convex optimization." 2025 IEEE 64th Conference on Decision and Control (CDC). IEEE, 2025.

Response: Per reviewer's suggestion, we have substantially broadened the discussion beyond the early decentralized OCO literature. The revision now covers accelerated gossip and graph-dependent global-loss regret, Erdős-Rényi networks, block communication, compressed messages and bit accounting, multiple consensus iterations for dynamic regret, unknown feedback delays, and recent improved bounds for compressed communication. We repeat the same positioning text below. The relevant added manuscript text (with its citations retained in the manuscript) is:

Decentralized OCO (D-OCO) with every agent active has previously been analyzed using prescribed step-size schedules. Recent global-loss D-OCO analyses make the dependence on topology and communication more explicit. Wan et al. use accelerated gossip to obtain nearly optimal dependence on the horizon, network size, and spectral gap, and their expanded journal work also develops projection-free variants with explicit communication-round guarantees. Lei et al. characterize regret and communication trade-offs over Erdős-Rényi random networks. Communication-efficient D-OCO has also been pursued through block updates with $\mathcal{O}(\sqrt{T})$ communication rounds, difference-compressed messages with explicit transmitted-bit accounting, and multiple primal and gradient consensus iterations for dynamic regret. More recent extensions address unknown feedback delays and compressed communication with improved graph- and compression-dependent regret and corresponding lower bounds. These works establish communication efficiency and graph-dependent regret as central concerns in D-OCO, but address different angles from the learning-rate-free, comparator-adaptive guarantee studied here. Comparator-adaptive distributed learning has also been developed for a protocol in which one agent is activated per round, while stochastic agent availability has been treated separately. These protocols differ from ours, where every agent predicts and incurs a local loss in every round; to the best of our knowledge, no parameter-free decentralized approach with network-regret guarantees has been proposed for this simultaneous protocol.

Parameter-free decentralization itself is not unique to our work. D-NASA targets stationarity for nonconvex stochastic objectives (Li et al.), Kuruzov and coauthors obtain linear convergence for strongly convex smooth objectives, and Chen and coauthors address composite convex objectives. These methods solve fixed-objective optimization problems and measure stationarity or objective convergence, not online optimization. Separately, DADAM is a decentralized adaptive-moment method for online optimization with a dynamic-regret analysis (Nazari et al.), but it retains optimizer hyperparameters. Our contribution instead concerns sequential predict-before-feedback losses: we present two coin-betting algorithm variants, both of which have comparator-adaptive average-local-regret guarantees, and one of which has a communication-dependent network-regret guarantee without a learning-rate scale.

Communication cost can instead be reduced through block updates, or measured at the bit level under compression; repeated consensus iterations also appear in dynamic-regret analyses. Our experiment varies rounds of full-precision gossip, so its scalar accounting is complementary to, rather than a substitute for, compressed-bit accounting.

Comment 7

While the paper emphasizes the advantage of being parameter-free, this benefit is restricted to convex online learning settings. In practice, many modern decentralized learning problems of interest are inherently nonconvex. However, the paper does not discuss how this limitation affects the significance of the proposed method. This restricted applicability should be more clearly acknowledged and justified.

Response: We agree that our guarantees do not cover nonconvex decentralized learning. The current manuscript states the scope directly as unconstrained convex OCO on a real Hilbert space and explains the associated linearized OLO reduction; it makes no nonconvex stationarity or convergence claim. This scope also distinguishes our setting from the fixed-objective nonconvex work discussed in response to the preceding concern. Adapting parameter-free (comparator-adaptive) online learning methods to nonconvex optimization is a well-known open problem, and far beyond the scope of this work. The manuscript now states:

In the rest of this work, we will assume we operate in the unbounded $\mathcal{X}$ case, specifically a real Hilbert space which we will denote by $\mathcal{H}$. We will also assume, without loss of generality, that the losses are 1-Lipschitz, since any L-Lipschitz loss can be rescaled to a 1-Lipschitz loss by dividing by L.
Our analysis applies to unconstrained OCO over the real Hilbert space $\mathcal{H}$. A standard technique for analyzing OCO problems is to linearize the convex loss functions using subgradients. This allows us to bound the regret of the convex problem by analyzing an equivalent Online Linear Optimization (OLO) problem.

Comment 8

While DECO-ii has a practical advantage over DECO-i in that it only gossips the accumulated gradient and no longer requires communicating the wealth variable, the paper does not sufficiently justify why the original communication design in DECO-i should be considered appropriate. In many decentralized learning methods (e.g., [4]-[6]), communication is typically performed over a single optimization-related state, such as the model or gradient. In contrast, DECO-i requires transmitting both the accumulated gradient and the wealth variable at each round. The paper should therefore clarify why this additional communicated variable is necessary and why this design servers as an appropriate reference point for comparison. From this perspective, DECO-ii appears less like a fundamentally new algorithm and more like a reformulation of DECO-i that eliminates the need for two-variable communication by incorporating the role of wealth into the betting-function-based update. In addition, because DECO-i communicates both wealth and accumulated gradients, whereas DECO-ii communicates only the accumulated gradients, the two methods do not operate under the same communication overhead per round. Therefore, the empirical comparison between DECO-i and DECO-ii is not fully fair unless the communication cost is properly normalized. The authors should explicitly justify this communication design and include experiments comparing the methods under the same communication budget, so that the reported performance differences can be attributed to the update rule itself rather than to unequal communication overhead.

[4] H. Xing, O. Simeone and S. Bi, "Federated Learning Over Wireless Device-to-Device Networks: Algorithms and Convergence Analysis," in IEEE Journal on Selected Areas in Communications, vol. 39, no. 12, pp. 3723–3741, Dec. 2021.
[5] Y. Wang, Y. Xu, Q. Shi and T. -H. Chang, "Quantized Federated Learning Under Transmission Delay and Outage Constraints," in IEEE Journal on Selected Areas in Communications, vol. 40, no. 1, pp. 323–341, Jan. 2022.
[6] Z. Zhai, X. Yuan, X. Wang and G. Y. Li, "Decentralized Federated Learning With Distributed Aggregation Weight Optimization," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 48, no. 3, pp. 3899–3910, March 2026.

Response: In response to earlier questions about the relation between DECO-i and DECO-ii (Comments 2 and 3), we clarified the communicated state and per-message payload. In particular, DECO-i is the direct decentralized form of classical wealth-based coin betting, whereas DECO-ii is the one-state betting-function variant that avoids communicating wealth. Nevertheless, both algorithms are our contribution. We compare the variants under the same graph, data order, horizon, and gossip schedule on all four LIBSVM streams, so their mixing operators are identical. Each directed transmission contains $d+1$ scalars for DECO-i and d for DECO-ii, a relative overhead of $1/d$ for DECO-i: 10% in the $d = 10$ synthetic experiment and 8.3%–16.7% across the real-data dimensions. Exact scalar equality would require different integer numbers or timings of gossip rounds and would therefore change the mixing operator. Because the payloads differ by only one scalar and the manuscript already reports the exact per-message difference under the matched schedule, we do not add a separate equal-budget experiment. Note that computing the wealth variable is often trivial, but computing the betting function can be more expensive, so the two variants are not necessarily equal in computational cost. The relevant added manuscript text is:

> DECO-**i** is the direct decentralized analogue of classical wealth-based coin betting: because its decision depends on both $\mathrm{Wealth}_{n,t-1}$ and $G_{n,t-1}$, both states are gossiped. DECO-**ii** is the one-state betting-function variant $x_{n,t} = h_t(G_{n,t-1})$ and gossips only G. Thus, DECO-**i** is a reference construction that preserves the classical wealth recursion, whereas DECO-**ii** reduces each directed-neighbor exchange from $d+1$ to d real scalars, where d is the dimension of the decision space $\mathcal{H}$.
>
> This adds one scalar to each d-dimensional message, a relative payload overhead of exactly $1/d$.
>
> The scalar count uses the same directed-neighbor accounting for every method: DECO-**i** sends (Wealth, G), while DECO-**ii**, DOGD, and the three adaptive baselines send one d-dimensional vector.